

Activity 2: Data Cleaning Challenge

Objective

Inspect the dataset and identify as many data quality issues as possible. For every issue you find, explain what is wrong and suggest the most appropriate data cleaning action.

Instructions

1. Read the dataset carefully.
2. Examine each row and column for data quality issues.
3. Write each issue you find on the answer sheet.
4. For every issue, indicate:
 - Type of issue (e.g., Missing Value, Duplicate, Outlier, etc.)
 - Location (Student ID or Row Number) (Also encircle the cell and put a number label)
 - Recommended solution (e.g., remove, correct, replace, investigate)
5. Complete the activity individually within the allotted time.

The first item is done for you.

Student Performance Prediction Dataset

Student ID	Name	Age	Attendance (%)	Study Hours/Week	Final Grade
1001	John	20	95	12	A
1002	Maria	19	88	10	B
1003	Kevin	?	91	9	B
1004	Anna	18	110	7	A
1005	Rose	eighteen ^{1.}	89	6	C
1006	John	20	95	12	A
1007	Carlo	21	75	-4	D
1008	Mia	20	NULL	11	B
1009	Alex	19	87	1000	A
1010	Sophia	19	82	8	Excellent
1011	Ethan	20	93	13	A
1012	Chloe	18	85	7	B
1013	Daniel	22	90	9	B
1014	Grace	19	81	5	C
1015	Liam	20	0	12	A

1016	Noah	17	96	14	A
1017	Ava	19	92	NULL	B
1018	Ella	20	89	10	b
1019	Lucas	20	ninety	8	B
1020	Olivia	18	84	6	C
1021	James	21	97	15	A
1022	Emma	-2	90	10	B
1023	Benjamin	19	87	8	Pass
1024	Isabella	20	93	11	A
1025	Mason	20	94	13	A
1026	Harper	19	101	9	B
1027	Elijah	20	86	five	C
1028	Charlotte	18	88	7	C
1029	Amelia	18	88	7	C
1030	Amelia	18	88	7	C

Answers:

1. Wrong data type used for the Age column. It showed **"eighteen"** at Student #1005. Solution: correct the data to use proper datatype: **18**.
2. **"?"** No input at Age column at Student #1003. Solution: Re-collect data
3. **"NULL"** No data at Attendance column at Student #1008. Solution: Re-collect data
4. **"110%"** Data exceeds normal threshold at Attendance Column Student #1004. Solution: Re-Check Data
5. **"- 4"** Improper Data at Study Hours Column, Student #1007. Solution: Re-check or Retract Data
6. **"1000"** Data exceeds normal threshold at Study Hours Column, Student #1009. Solution: Re-evaluate or Retract Data
7. **"Excelent"** Wrong placeholder inputted at Final Grade Column, Student#1010. Solution: Change placeholder
8. **"-2"** Improper Data inputted at Age Column, Student#1022.Solution: Remove/Re-evaluate Data
9. **"Duplicated Row"** Student#1029 & Student#1030. Solution: Remove Row
10. **"Ninety"** Wrong Data type used at Age Column, Student#1019. Solution: Correct Data to proper datatype